PROJECT 18 · SEMESTER 5 APPLIED ML & DEEP LEARNING

Surface Water Quality Prediction

Estimating the CCME Water Quality Index with gradient-tree ensembles, aquatic-chemistry features and a dual-head neural network — and an honest audit of what the scores mean



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2.82 M observations · 5 countries · 1940–2023

THE PROBLEM

Lab assays are slow and sparse; an index estimator could triage faster

Grab sample

Field visit, discrete in space and time

→

Lab assay

BOD5 needs 5 days (120 h) incubation

→

Index

Composite score computed after results arrive

Project objective

Learn a fast surrogate: measured physico-chemical parameters (pH, temperature, DO, BOD, ammonia, orthophosphate, nitrogen, nitrate) → CCME WQI score and quality tier.

Scope of claims

- Demonstrated: offline regression and classification on a historical dataset
- Not demonstrated: real-time sensing, field deployment, replacing laboratory testing

OBJECTIVE AND TARGET

The target is the CCME Water Quality Index: 100 minus a distance from ideal

$$\text{CCME WQI} = 100 - \sqrt{(F_1^2 + F_2^2 + F_3^2)} \,/\, 1.732$$

F_1 Scope

% of variables that fail their guideline

F_2 Frequency

% of individual tests that fail

F_3 Amplitude

Size of excursions: nse / (0.01·nse + 0.01)

Five quality tiers (share of dataset rows)

Poor 0–44	Marginal 45–64	Fair 65–79	Good 80–94	Excellent 95–100
4.5%	11.6%	11.3%	36.4%	36.2%

Index values in the data: mean 85.05, median 90.64, range 32.15–100 (strongly left-skewed). Tier shares from the technical report; they sum to ≈100%.

DATASET SCALE AND COMPOSITION

2.82 million observations, but 75% of them come from England

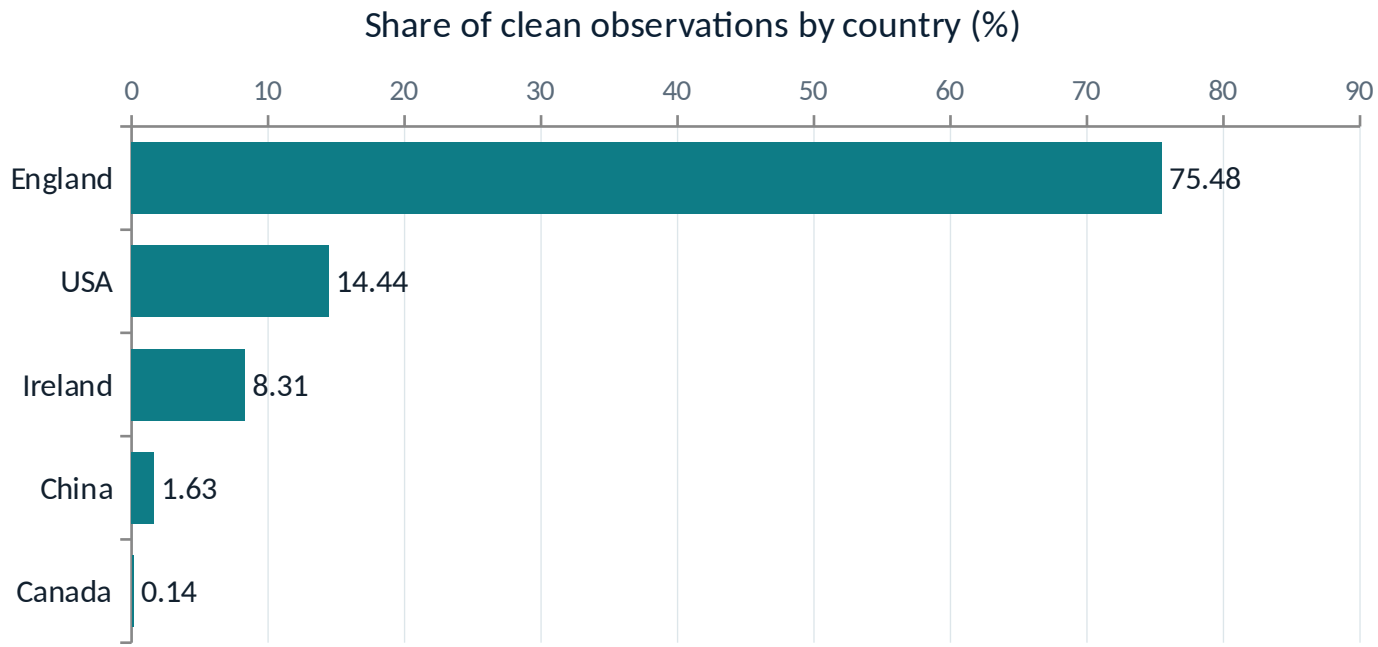
2,827,977
raw records

2,819,617
clean records

5
countries

1940–2023
83 years

38
model input features



Clean row counts

England 2,128,128
USA 407,138
Ireland 234,409
China 45,996
Canada 3,946

Rivers are 65% of waterbodies. Results mainly reflect English rivers; Canada and China are too small to support country-level claims.

DATA AUDIT AND PHYSICAL-BOUND SCREENING

Zero nulls — but 8,360 physically impossible readings were removed



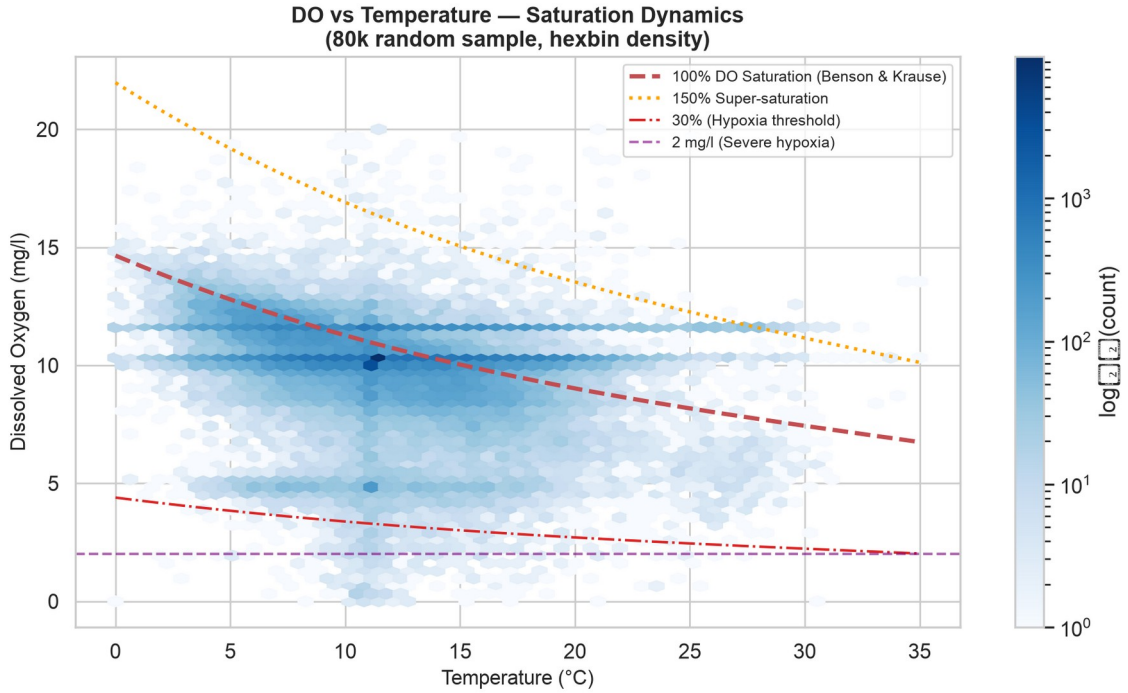
Parameter	Valid range	Violations
pH	3.0 – 12.0	1,581
Temperature (°C)	0 – 45	6,850
Dissolved O_2 (mg/L)	0 – 25	0
Ammonia / BOD (mg/L)	≥ 0	2 / 2
Orthophosphate (mg/L)	≥ 0	10
Nitrogen / nitrate	≥ 0	0 / 0

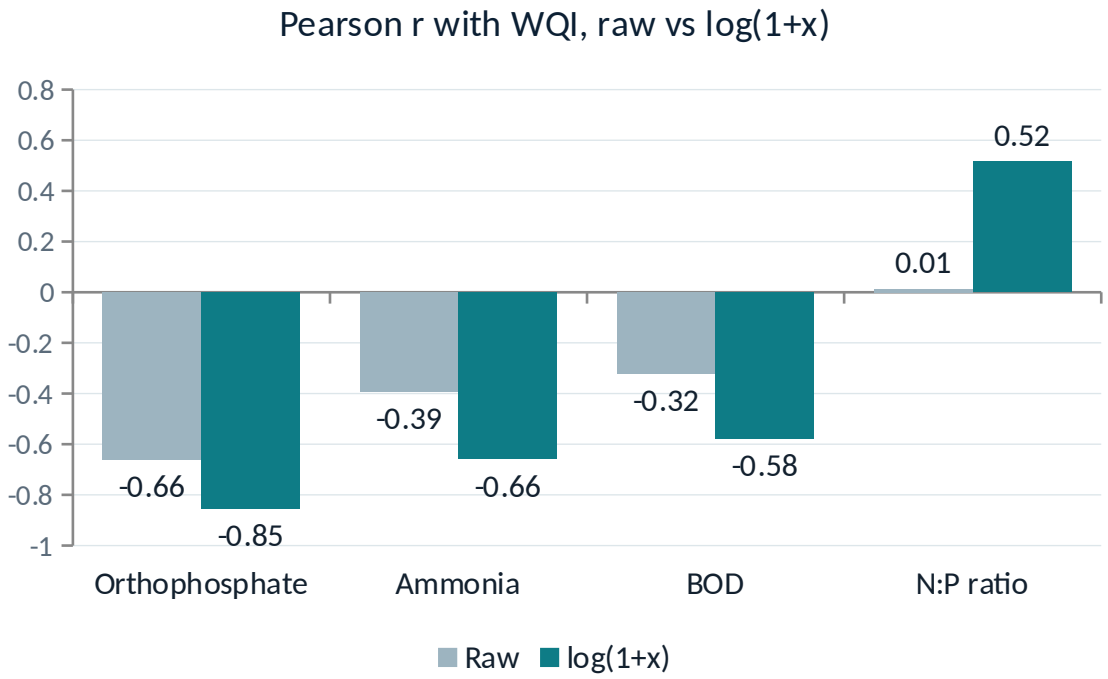
Counts overlap; the 8,360 is the de-duplicated union of violating rows.

Missing vs implausible. The legacy claim of ~35% missing values was disproved: 0 nulls across the 14 source columns, so no imputation was used. Screening removes sensor artefacts (e.g. negative concentrations), not gaps.

EXPLORATORY FINDINGS

Oxygen follows temperature; nutrients are heavily skewed, and log(1+x) straightens them





DO vs temperature (80k sample): saturation falls as water warms (Henry's law, Benson–Krause). Source: project report Fig. 2.

Qualification A better Pearson r is not better out-of-sample prediction. log(1+x) is monotonic: rank correlation is unchanged and tree splits are invariant to it.

DOMAIN-INFORMED FEATURE ENGINEERING

Eight measurements become 38 features, grouped by the chemistry they encode

8

Base sensors

pH, temperature, DO, BOD, ammonia, orthophosphate, nitrogen, nitrate

7

Aquatic stoichiometry

Benson–Krause DO saturation/deficit, Redfield N:P, Emerson un-ionized NH_3, BOD:DO

7

log(1+x) variants

Compress nutrient skew spanning orders of magnitude

6

Station memory

Shift-1 lags and rolling-3 means, sorted by Area and Date; backward-looking only

7

Temporal

year, month, quarter, sin/cos of month and day-of-year

2

Context

country code, waterbody-type code

1

exceedance_counter

Count of guideline violations across parameters

Total: 8+7+7+6+7+2+1 = 38 features

Station lag Spearman ρ(hist_ortho_lag1, WQI) = −0.78

PIPELINE ARCHITECTURE AND TESTING

Features are built before the split; scaling is fitted on training data only

1 Ingest & audit
5 countries, 0 nulls

2 Physical bounds
−8,360 rows

3 38 features
incl. station lags

4 Target quarantine
3 columns named in tests

5 Stratified 70/15/15
random, by WQI tier

6 Scale on train
StandardScaler (MLP)

7 Train & evaluate
RF · XGB · MLP

8 TreeSHAP
XGBoost, validation

Navy = before split · Amber = split/leakage-sensitive · Teal = fitted or evaluated after split

Automated tests (pytest)

12 tests in 4 modules: features 4 · models 3 · leakage 2 · metrics 3.
Reported 100% pass in 1.65 s (not re-run in this review).

They check: bound filtering, no look-ahead in lags, named target columns absent from X, metric maths, checkpoint loading.
They do not show: that features are free of target-derived information or that scores generalise.

MODELING STRATEGY AND EVALUATION DESIGN

Four regressors, one random stratified split, and conflicting hyper-parameter records



Random, class-stratified by WQI tier · no station or time hold-out · reported metrics are on validation; test results are not reported

Mean baseline

Predicts the training mean (85.05). The floor any model must beat.

Random Forest

Bagged, de-correlated trees; max_features = √p.

Config conflict: 100 trees/depth 20 vs 30 trees/depth 14

XGBoost

Boosted trees, squared-error objective.

Config conflict: 300 rounds/η 0.05/depth 8 vs 150/η 0.08/depth 6

Dual-head MLP

38→256→128→64, then regression head (Smooth-L1) and 5-class head (weighted CE); AdamW, 8 epochs.

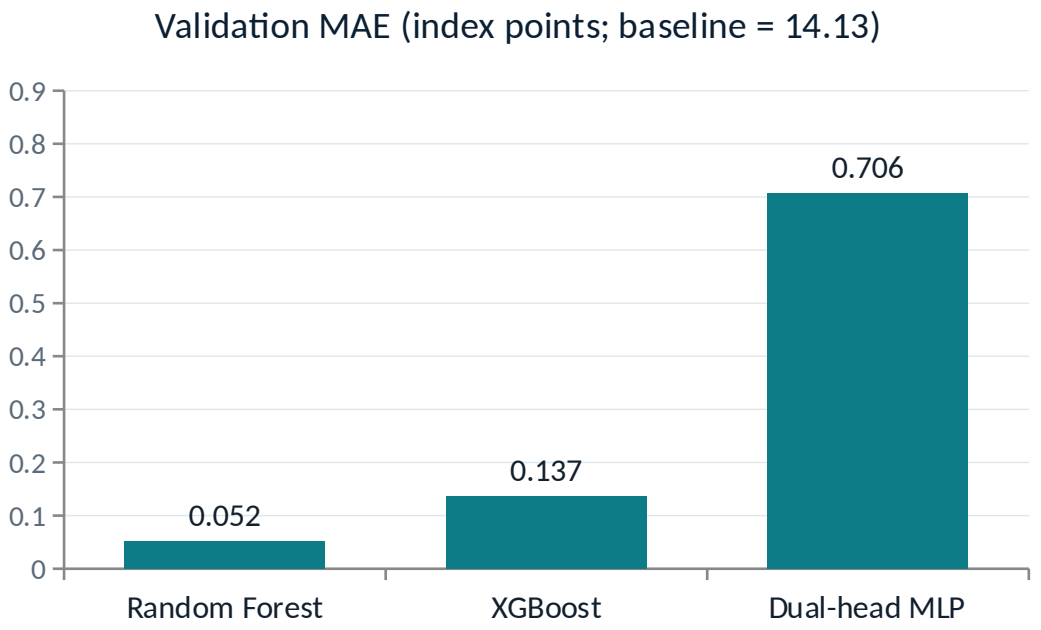
Reported in repo technical report

MODEL BENCHMARKING

Tree ensembles reproduce the index almost exactly; the MLP is close but less precise

Model	MAE	RMSE	R^2
Mean baseline	14.130	17.656	0.0000
Random Forest	**0.052**	**0.208**	**0.9999**
XGBoost	0.137	0.374	0.9996
Dual-head MLP	0.706	1.156	0.9957

MLP also classifies the 5 tiers: accuracy 96.09%, macro-F1 0.9485, Brier 0.0533, ECE 1.27% (validation).



Qualification These scores measure how well each model re-computes a deterministic formula on randomly held-out rows. They are not evidence of independent forecasting skill (slide 11). Latency claims omitted: provenance unverified.

RELIABILITY AND VALIDITY

Why $R^2 \approx 0.9999$ needs careful reading: three open audit findings

1 Deterministic target

The index is computed from the same eight measurements used as inputs, against fixed guideline limits. The team's viva guide calls it an "exact deterministic mathematical aggregation". High R^2 = the model learned the formula.

Confirmed from docs

2 exceedance_counter

A guideline-violation count sits in the 38 inputs. It uses the same exceedance information that builds F_1. Tests name only 3 quarantined target columns. Source code could not be read.

Unresolved risk

3 Random split

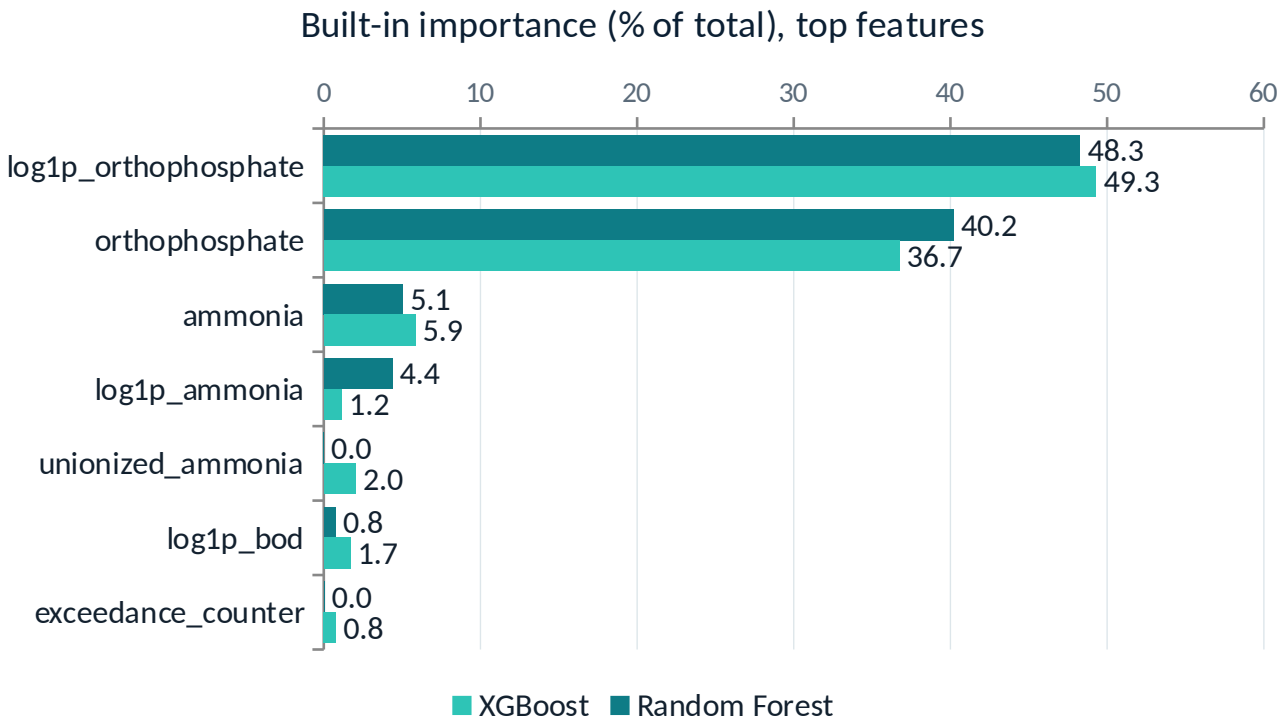
Rows are split at random within WQI tiers. Station lags are built before splitting, so a row's neighbours can sit in training. No station-held-out or time-forward check; test set unreported.

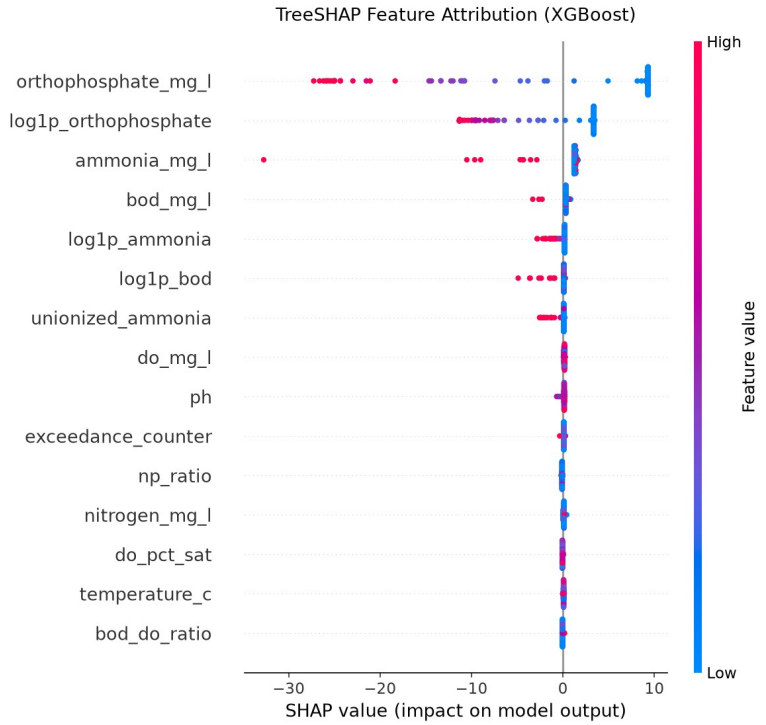
Confirmed from docs

Can claim
Fast, accurate surrogate of the CCME computation on this dataset (validation).

Cannot claim
Leakage-free generalisation to new stations or future years; independent water-quality forecasting.

Orthophosphate dominates the model — a property of the index and data, not a causal finding





TreeSHAP beeswarm, XGBoost (project report Fig. 5)

Qualification Importance describes the model, not the river. Orthophosphate's share is 88% (RF) / 86% (XGB) of gain/impurity importance, not "88% of splits".

LIMITATIONS, STATUS AND NEXT STEPS

Phase 1 is complete; the deep-learning model exists, and stricter validation comes next

Confirmed from documents

- Country imbalance: England ≈75% of rows
- Random split; no station/time hold-out
- Validation metrics only; test set unreported
- Report and repo disagree on tree settings, country counts, importance wording

Open / not verified

- Whether exceedance_counter inflates scores
- Latency figures (hardware, batch size)
- Duplicates and cross-partition near-duplicates
- Source code and notebooks not reviewed

Planned (not done)

- Ablation without exceedance_counter
- Station-held-out and time-forward splits
- Report held-out test scores
- External validation; uncertainty beyond MLP calibration

Implemented (validation only): dual-head MLP — R^2 0.9957, MAE 0.706, accuracy 96.09%, ECE 1.27% — per the repository technical report.

A reproducible, well-tested baseline — with a clear route to a stricter evaluation

Built: audited 2.82 M-row dataset, 38 chemistry-driven features, RF/XGBoost/MLP benchmarks, SHAP, 12 tests.

Strongest finding: models re-compute the CCME index with validation MAE ≤ 0.14 (trees).

Key caveat: the target is a formula of the inputs and the split is random, so run leakage-free ablations next.

Team and roles

Mayank Sharma – EDA & repository management lead
Aman Soni – model training lead (baseline, RF, XGBoost)
Aaryan Gera – data preprocessing (cleaning, dataset preparation)
Chaitanya Kumar – feature engineering & pipeline
Ankit Kumar Pandey – model evaluation & validation (MAE, RMSE, R^2)
Pranav – research, documentation & presentation

Key references

Breiman (2001) Random Forests. Machine Learning 45(1)
Chen & Guestrin (2016) XGBoost. KDD
Lundberg & Lee (2017) SHAP. NeurIPS
CCME (2001) Water Quality Index 1.0 Technical Report
Benson & Krause (1984) Limnol. Oceanogr. 29(3)
Redfield (1958) Am. Scientist 46(3)
Emerson et al. (1975) J. Fish. Res. Board Can. 32(12)

Data: national agencies as named in the project report (England EA, USGS, EPA Ireland, MEE China, ECCC). Code: github.com/MAYANKSHARMA01010/Semester_5_AML_DL